



Effects of integrated takeover request warning with personal portable device on takeover time and post-takeover performance in level 3 driving automation

Niloy Talukder^a, Chris Lee^{a,*}, Yong Hoon Kim^a, Balakumar Balasingam^b, Francesco Biondi^c, Aditya Subramani Murugan^d, Eunsik Kim^d

^a Department of Civil and Environmental Engineering, University of Windsor, ON N9B 3P4, Canada

^b Department of Electrical and Computer Engineering, University of Windsor, ON N9B 3P4, Canada

^c Department of Kinesiology, University of Windsor, ON N9B 3P4, Canada

^d Department of Mechanical, Automotive & Materials Engineering, University of Windsor, ON N9B 3P4, Canada

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ABSTRACT

Introduction: Level 3 driving automation defined by the Society of Automotive Engineers (SAE) requests human drivers to drive manually when the vehicle cannot perform the driving task. In this regard, researchers have studied the integrated takeover request (TOR) which provides visual and auditory TOR warning in both vehicle interface (e.g., dashboard, windshield (head-up display) and personal portable device (PPD) (e.g., cell phone, tablet). However, these studies neither used auditory TOR warning in PPD nor examined the effect of use of headphone on takeover. Thus, this study evaluates the effects of the integrated TOR with the use of headphones on the takeover time and the post-takeover performance. **Method:** The behavior of 60 drivers was observed in the driving simulator experiment. During the experiment, the drivers watched a video on a tablet in automated driving, received the TOR warning, and manually drove in the lane change and pullover scenarios. The survey was also conducted to ask drivers' experience and preference for TOR warning. **Results:** The integrated TOR significantly reduced the takeover time compared to the conventional TOR which provides the TOR warning in vehicle interface only. The integrated TOR also improved the post-takeover performance as indicated by more stable steering operation and safer driving behavior after TOR warning. However, the use of headphones did not significantly reduce the takeover time or improve the post-takeover performance for the integrated TOR. The participants generally perceived higher subjective comfort and safety level with the integrated TOR than the conventional TOR. **Conclusions:** The integrated TOR with auditory warning in PPD can significantly reduce the takeover time and improve the post-takeover performance in both urgent and less urgent conditions. **Practical applications:** The integrated TOR with auditory warning in PPD can be applied to SAE Level 3 driving automation for safe transition from automated to manual driving.

1. Introduction

The number of deaths by traffic accidents reached 1.35 million annually according to the Global status report on road safety (World Health Organization, 2019). In this regard, autonomous vehicles (AV) and advanced driver assistance systems (ADAS) could decrease traffic deaths by 90% and save millions of lives and billions of dollars every year in health care costs (McKinsey & Company, 2015). As the demand for AVs is expected to grow, the leading car manufacturers have been

developing AVs with Level 3 driving automation defined by Society of Automotive Engineers (SAE), which will allow drivers to be free from the primary task of driving and engaged in the secondary tasks while traveling.

However, it's possible that Level 3 AVs cannot perform the driving task and operate safely (e.g., sensors cannot detect objects in poor visibility condition). In this case, AVs generate the takeover request (TOR) so that drivers can take over the driving task and manually drive the vehicle. The time for transition from automated to manual driving or the

* Corresponding author.

E-mail addresses: talukdern@uwindsor.ca (N. Talukder), cclee@uwindsor.ca (C. Lee), kim523@uwindsor.ca (Y.H. Kim), singam@uwindsor.ca (B. Balasingam), fbiondi@uwindsor.ca (F. Biondi), murugan6@uwindsor.ca (A.S. Murugan), eskim@uwindsor.ca (E. Kim).

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takeover time is important to ensure driver safety. The takeover is defined as the combination of physical, visual, and cognitive readiness after TOR is initiated (Zeeb et al., 2015). In most recent studies, the start time of takeover is considered as the time of 10% depression of brake pedal or more than 2° rotation of steering wheel by the driver (Zhang et al., 2019; Louw et al., 2017; Zeeb et al., 2015). Post-takeover performance has been evaluated based on time to collision (TTC), acceleration, lane change duration, and time headway to the lead vehicle during the takeover time (Louw et al., 2015; Bueno et al., 2016; Wan & Wu, 2018). Typically, the drivers who showed safer driving behaviors after the TOR warning was generated had longer TTC, lower acceleration, shorter lane change duration, and longer time headway. Thus, these parameters reflect better post-takeover performance.

Secondary tasks and modality of TOR significantly affect the takeover time and the post-takeover control (McDonald et al., 2019). Handheld tasks involving personal portable devices (PPD) such as cell-phones and tablets will be common secondary tasks during automated driving. However, these handheld secondary tasks have higher adverse effect on both takeover time (Wan & Wu, 2018; Wandtner et al., 2018a) and post-takeover performance (Zeeb et al., 2017; Wandtner et al., 2018a) than non-handheld secondary tasks. This can be because of longer required time for physical and visual readiness (Zhang et al., 2019; Radlmayr et al., 2014; Wan & Wu, 2018; Wandtner et al., 2018a; Zeeb et al., 2017).

To reduce the takeover time when drivers use PPD for secondary tasks, the TOR warning can also be provided in PPD in addition to the vehicle interface. In this study, the combination of TOR warning from both PPD and vehicle interface is called “the integrated TOR.” The integrated TOR can display the visual warning to drivers while facilitating their takeover time since the gaze of the driver is already on the PPD screen during the secondary task. In addition to the reduced gaze redirection time from visual and auditory warning, using PPD can also reduce the delay caused by the secondary task. Moreover, listening to auditory warning from PPD using headphone can potentially reduce the time for TOR recognition and visual and physical readiness. Although headphone use should not be permitted for Level 3 driving automation (Gerber et al., 2023), it is worthwhile to test the effects of headphone use on drivers’ response to the integrated TOR as a hypothetical method of improving the auditory warning from PPD.

The past studies demonstrated that the integrated TOR warning using PPD can effectively reduce the takeover time and improve the post-takeover performance, particularly when people are engaged in the secondary task using PPD (Melcher et al., 2015; Miller et al., 2015; Politis et al., 2017; Yoon et al., 2019; White et al., 2019). However, these studies neither used auditory TOR warning in PPD nor examined the effects of use of headphone and visual warning that interrupts the video on PPD on takeover. Thus, more studies are needed to evaluate the effectiveness of multimodal (visual and auditory) TOR warning using PPD in reducing the takeover time and improving the post-takeover performance.

Thus, the objectives of this study are: (1) to investigate the effectiveness of the integrated TOR in reducing the takeover time and improving the post-takeover performance compared to the conventional TOR that provides the TOR warning in vehicle interface only; and (2) to evaluate the effects of use of headphones during the secondary task on the takeover time and the post-takeover performance. This study will test the following hypotheses that have not been tested in previous research:

Hypothesis 1.1: The integrated TOR reduces the time for visual readiness (i.e., the time it takes to check the TOR warning) compared to the conventional TOR.

Hypothesis 1.2: The integrated TOR reduces the time for physical readiness (i.e., the time it takes to be physically ready for manual control) compared to the conventional TOR.

Hypothesis 2.1: The auditory warning through the headphone from PPD and the visual warning that interrupts the video on PPD reduce the take-over time compared to the auditory warning from vehicle interface.

Hypothesis 2.2: The auditory warning through the headphone from PPD and the visual warning that interrupts the video on PPD improve the post-takeover performance compared to the auditory warning from vehicle interface.

Hypothesis 3: The integrated TOR reduces the drivers’ mental workload.

Hypotheses 2.1 and 2.2 rely on the postulate that auditory warning through the headphone from PPD will improve salience of warnings by blocking ambient sounds and reducing drivers’ cognitive distractions. Hypothesis 3 relies on the postulate that the integrated TOR will increase the drivers’ decision-making ability and level of comfort during the takeover process after the TOR is generated.

2. Literature review

2.1. Effects of secondary task on takeover time and post-takeover performance

The driver’s ability of re-engaging in the driving task depends on the secondary task they perform during autonomous driving. The visual, auditory, physical, and cognitive demands of secondary task affect the efficiency and safety during the vehicle’s transition of control (Zeeb et al., 2016; Marberger et al., 2017). In general, the secondary task significantly increases the driver’s takeover time (Wandtner et al., 2018b; Zeeb et al., 2017; Zhang et al., 2019) and negatively affects the post-takeover performance. On the other hand, the drivers who take longer time to view the driving scene have higher situation awareness to take over (Liang et al., 2021).

When drivers are involved in the secondary tasks, it adversely affects the longitudinal post-takeover control and increases chances of crash in higher traffic density (Radlmayr et al., 2014) and decreases the minimum TTC (Gold et al., 2016; Korber et al., 2016). The secondary tasks also affect the post-takeover control, which results in higher lateral acceleration (Louw et al. 2015), higher average and standard deviation of lane position, the lane exceedance (Wandtner et al., 2018b; Zeeb et al., 2016), and longer time and less stability of steering operation (Pan, He, Wang, Cheng, & Dai, 2023; Wu et al., 2020; Zhang, Fard, Davy, Parida, & Robinson, 2023). Furthermore, the secondary tasks increase the time required to change lanes and maximum steering wheel angle (Bueno et al., 2016).

In particular, the handheld secondary task using PPD (e.g., phone conversation or trivia game) significantly increases the takeover time (Wan & Wu, 2018). This is mainly because when switching from the task using handheld PPD to take over, the driver takes extra time to decide where to put down the device (Wandtner et al., 2018a; Zeeb et al., 2017). Wan and Wu (2018) showed that watching a video and reading using PPD results in longer reaction time, higher standard deviation of reaction time, and lower TTC relative to other tasks during automated driving. Shi and Bengler (2022) found that the handheld secondary tasks with higher cognitive demand such as playing Tetris on a tablet increased takeover time compared to the task with lower cognitive demand such as watching a video on a laptop.

The secondary task using PPD also has negative impacts on post-takeover control. Delayed visual and manual reaction due to the task using PPD causes the driver’s urgent evasive maneuvers instead of a controlled action (Zeeb et al., 2017). Zeeb et al. (2017) also suggested that when PPD was held in hand instead of being mounted, post-takeover performance was degraded.

2.2. Modalities of takeover request

Different modalities of TOR provided to the drivers greatly influence the takeover time, performance, and quality of the takeover (Naujoks et al., 2014; Petermeijer et al., 2017). Several studies have investigated different alert methods, information provided for successful takeover and types of modality (auditory, visual, vibrotactile, and combination of these modalities). The most common type of modality was the combination of visual and auditory warning (Eriksson et al., 2017; Gold et al., 2015; Miller et al., 2015; Melcher et al., 2015). Naujoks et al. (2014) and Politis et al. (2015) found that multimodal cues are more efficient. Petermeijer et al. (2017) showed that multimodal warning resulted in faster steer-touch time and better usefulness rating and satisfaction among the users. Bazilinskyy et al. (2018) also reported that participants preferred the multimodal TOR warning compared to the unimodal TOR warning. Yun and Yang (2020) found that the multimodal TOR warning showed shorter drivers' reaction time for TOR and better post-takeover performance than the unimodal (visual only) TOR warning.

Different interfaces of the visual warnings were also examined in the previous studies – a mounted screen or a vehicle display (Zeeb et al., 2017; Kaye et al., 2021), a dashboard (Yun & Yang, 2020; Wu et al., 2019; Melcher et al., 2015), a tablet (Yoon et al., 2019; Miller et al., 2015; Politis et al., 2017), and a cellphone (Melcher et al., 2015).

In general, audio alerts were generated from the vehicle speakers in the previous studies (Eriksson et al., 2019; Korber et al., 2016; Melcher et al., 2015; Gold et al., 2015; Gold et al., 2016). However, if the driver is involved in the secondary task using PPD and uses a headphone, they may not be able to hear the audio alert from the vehicle speaker.

2.3. Integrated takeover request warning using personal portable device

During the secondary task using PPD, the audio alert from PPD can help reduce the drivers' takeover time because they can see the warning more quickly and better comprehend the warning message while the secondary task is stopped. For instance, Melcher et al. (2015) integrated the cellphone with the on-screen TOR message in the dashboard by providing the same TOR message on the participants' cellphone. They compared the participants' reaction times between the integrated TOR using the cellphone and the TOR without integration of the cellphone (i. e., the on-screen TOR only) using a driving simulator. The participants were asked to perform an artificial secondary task (quiz game) using the cellphone and multimodal (audio and visual) TOR warning was provided for the driving simulator experiment. In this integrated TOR warning, the cellphone only provided the visual alert while the simulator provided the auditory alert. The study suggested that providing 10 s to take over the driving task was sufficient and the integrated TOR warning with the cellphone reduced the mean takeover time.

Politis et al. (2017) also conducted a similar study by integrating the TOR using a tablet and playing games as the secondary task. The study provided the visual warning from the tablet but auditory alert using an external speaker. However, the tablet was placed on a stand and the participants did not hold the tablet in their hands. The study found that this integrated TOR warning reduced the takeover time and the lateral deviation.

Miller et al. (2015) investigated the effects of different secondary tasks (supervision and media consumption) on the takeover time and the post-takeover performance when the integrated TOR warning was provided in a tablet. They found that there was no significant difference in the drivers' reaction time and minimum distance headway between supervision and media consumption tasks. In contrast, Yoon et al. (2019) found that different secondary tasks affected the takeover time with the integrated TOR in a tablet. White et al. (2019) observed that the two-stage integrated TOR warning messages in a tablet – “Check for hazards” followed by “Resume control” – increased the drivers' situational awareness before manual control compared to the single TOR warning message of “Resume control.” However, these studies did not

compare the integrated TOR with the conventional TOR. Moreover, similar to Politis et al. (2017), the tablet was located in a fixed position instead of being held by drivers in Yoon et al. (2019) and White et al. (2019).

3. Methodology

3.1. Equipment

To investigate the effects of the integrated TOR on takeover time and post-takeover performance, driver behavior was observed using a fixed-base miniSim™ driving simulator. The simulator consists of three LCD monitors placed within a horizontal field-of-view with two side-view mirrors and a rear-view mirror. Surrounding sound from the speakers and vibration from the driver seat enhance the sense of reality.

The auto-drive feature of the miniSim driving simulator (AutoDrive) is capable of longitudinal and lateral vehicle control when the automated driving is activated using the automation button. Automated driving is also capable of overtaking other vehicles and changing lanes. Once the driver moves the steering wheel or presses the brake pedal, the automated driving will stop. Automated driving can also be turned off using the automation button. The status of automated driving is displayed on the screen left to the steering wheel. A Samsung Galaxy S6 Lite tablet with a 10.4-inch display was provided to participants during the experiment. An Android application was developed for the TOR to be provided from the tablet. The tablet was used for the secondary task and the display of TOR warning in the scenarios with the integrated TOR as shown in Fig. 1.

In this study, the takeover time is defined as the time from when TOR warning is generated to when the driver is physically ready for manual control (e.g., put hands on steering wheel or put feet on brake/accelerator pedal). To better understand how the integrated TOR can reduce the takeover time, the takeover time was classified into the components of physical and visual readiness time. The takeover time consists of: (1) the TOR Recognition Time (the time it takes to recognize that the TOR warning is generated); (2) the TOR Gaze Redirection Time (the time it takes to check the TOR warning); (3) the Clearance Time (the time it takes to put away the PPD from hands); (4) the Maneuver Gaze Redirection Time (the time it takes to check surrounding environments before manual control); and (5) the Readiness Time (the time it takes to be physically ready for manual control).

To measure the physical readiness time for takeover, the times when the driver touched the steering wheel and brake/accelerator pedals were detected using the sensors. The Adafruit extra-long force-sensitive resistor was used for steering wheel and the square force-sensitive



Fig. 1. Participant involved in secondary task during automated driving.

resistor was used for brake and accelerator pedals as shown in Fig. 2. The Arduino board Adafruit Metro 328 was used to connect the force-sensitive resistors with a breadboard and wire connection.

Two video cameras were also installed to record right-side profile of the participant's body and the front view of their face for detecting their visual readiness. First, the 3-dimensional position of 33 landmarks on the participant's body in each frame of the side profile video was analyzed. Second, the angles of different body joints such as neck angle, right upper arm angle, right lower arm angle, right knee angle and right foot angle were calculated from the position data.

A similar Python program written with Mediapipe Toolkit's face mesh solution (Google/Mediapipe, 2019) was used to estimate the participants face orientation during the experiment using the video recorded from the front view camera. The Mediapipe face mesh solution is a computer vision tool used to estimate the 3D position of facial landmarks from 2D images or videos. The solution outputs a set of 468 landmarks that can be used to track facial movement and expression. For the purposes of this study, only six landmarks were selected (Fig. 3; nose tip, chin, left eye corner, right eye corner, left mouth corner, and right mouth corner) since they were sufficient to accurately estimate the orientation of the participant's face while reducing computational time. The Perspective-n-Point (PnP) problem is a fundamental task in computer vision that can be used in estimating the position and orientation of an object in 3D space from a 2D image. This study utilized PnP solution called SQPnP that is implemented by OpenCV, an open-source computer vision library, in the solvePNP method (OpenCV: Perspective-n-Point (PnP) Pose Computation, 2023) to estimate the orientation of the participant's face.

3.2. Measurement of the components of takeover time

The components of physical and visual readiness time were measured using the data from resistors and video cameras as follows. Since it was not possible to observe when they actually recognized the TOR warning, the TOR Recognition time could not be separately measured. Thus, the TOR Recognition Time was combined with the TOR Gaze Redirection Time for the conventional TOR and the Clearance Time for the integrated TOR. The video cameras recorded the time using Time Stamp Camera Android application. First, the time when TOR was generated (TOR time) in each scenario was recorded from the videos. As shown in Fig. 4, the TOR Gaze Redirection Time (only for the conventional TOR) was measured from the TOR time to the time when facial landmarks indicated that the participant looked forward based on the reference landmark (initial position of participants who are watching a

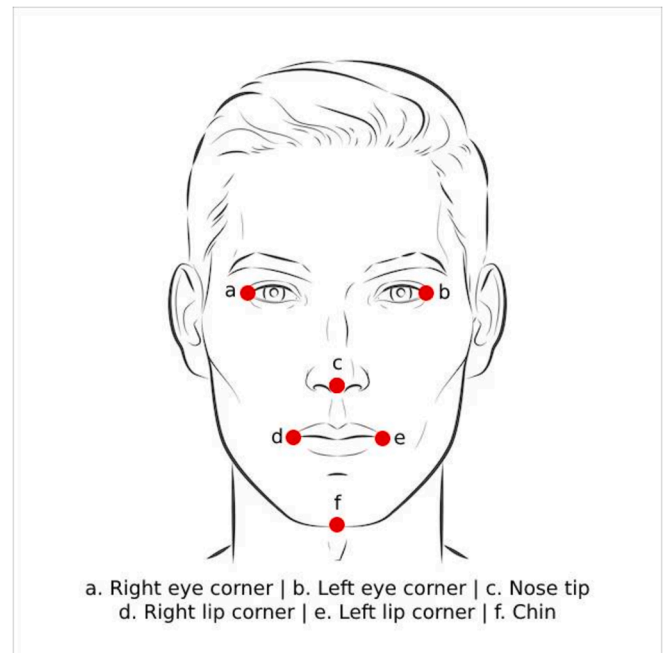


Fig. 3. The position of landmarks selected for face tracking.

video). The TOR Gaze Redirection Time was not measured for the integrated TOR because the participants received the TOR warning on the tablet and they did not need to look forward to check the TOR warning in the windshield. The Clearance Time ends when the participants put the tablet on their lap. This time was measured based on the lower right arm angle.

The Maneuver Gaze Redirection Time was also measured from the end of the Clearance Time to the time when the participants looked forward based on the reference landmark. The Readiness Time was measured from the end of the Maneuver Gaze Redirection Time to the time when the participants put their hands on the steering wheel or the time when they put their feet on the brake or accelerator pedal, whichever comes before. This end time of the Readiness Time was extracted from the force-sensitive resistors.

3.3. Driving scenarios and design of TOR

A total of 60 participants (38 males, 22 females) participated in the



Fig. 2. Force-sensitive resistors in steering wheel, brake pedal and accelerator.

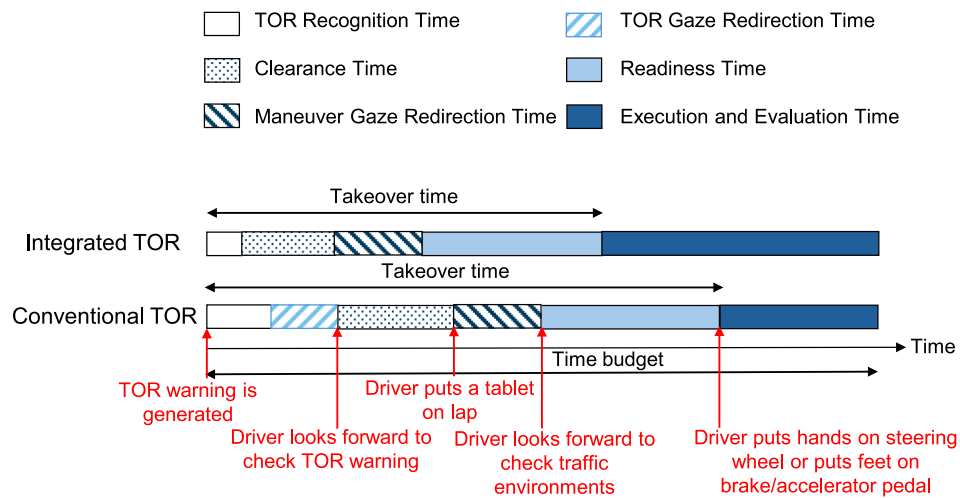
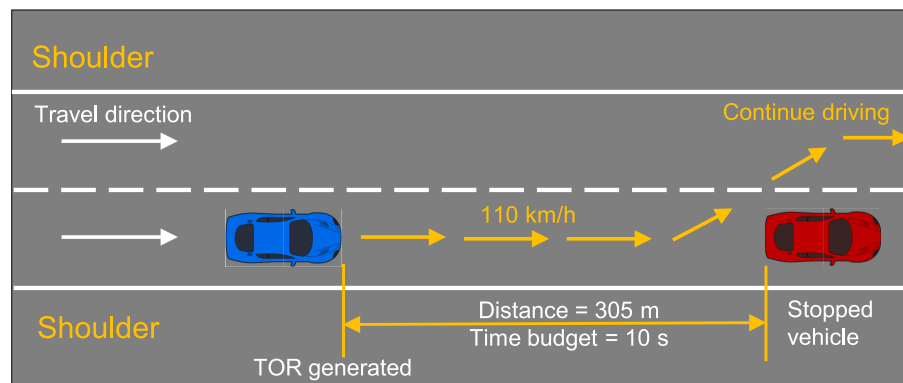


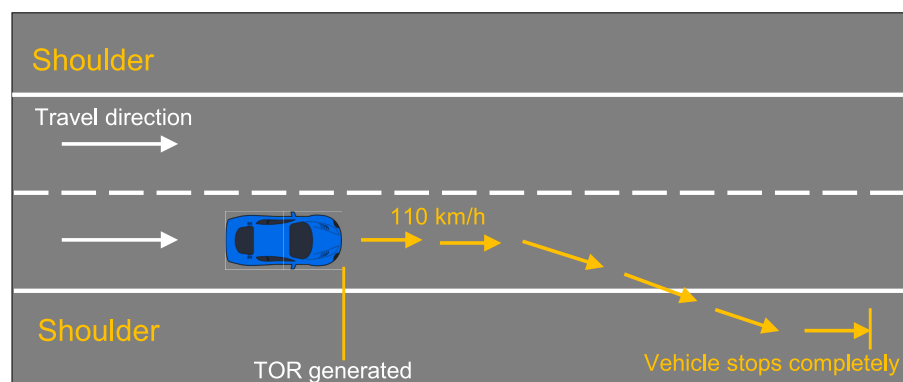
Fig. 4. Measurement of components of visual and physical readiness time.

experiment. Among them, 50 participants tested the driving simulator scenarios without resistors and video cameras in November 2021 due to unavailability of the equipment. Ten participants tested the scenarios with resistors and video cameras in February 2022. Since the past studies found that male drivers tend to react faster than female drivers (e.g., [Lipps et al., 2011](#)), potential differences in takeover reflections between male and female drivers were considered in this study. Their

ages varied from 21 to 50 years old (mean (M) = 28.4, standard deviation (SD) = 4.91). All participants were graduate and undergraduate students. A valid driving license with a minimum of 1 year of driving experience was required to participate. The participant's driving experience was also recorded (M = 5.22, SD = 3.8). None of the participants had prior experience with automated driving. Participation in the experiment was voluntary and the participants were compensated \$20



(a) Lane change scenario



(b) Pullover scenario

Fig. 5. Takeover scenarios for driving simulator experiment.

for the participation. The simulator experiment was cleared by the University of Windsor Research Ethics Board.

In the driving simulator experiment, the vehicle drove on a four-lane freeway (two lanes in each direction), which was a combination of straight and curved road in normal weather condition. Initially, participants activated automated driving by pressing the automation button. The preset design speed for the experiment was 110 km/hour and the vehicle drove in the right lane. During the automated driving, the participants were engaged in the secondary task (i.e., watch a video on a tablet).

Before the experiment, the participants were provided with a tablet with subscribed account of YouTube and Prime video from which they selected their preferred video to watch. The minimum duration of the video selected by the participants was 15 min. Thirty out of the 60 participants used an over ear headphone with noise canceling function (Sony WH-H900N/BM) while watching the video. The participants were informed that they were not required to monitor the driving environment during automated driving. This instruction was provided to the participants to have the participants focus more on the secondary task during the automated driving. In this situation, the effectiveness of the integrated TOR in reducing the takeover time can be better identified. Before the takeover situation, other vehicles simultaneously drove in the same direction as the subject vehicle. The subject vehicle completed the required evasive maneuvers (e.g., changing lanes or slowing down) by itself to safely continue driving.

During the automated driving, visual and audio TOR warning was generated in the following two circumstances: (1) the Lane change scenario and (2) the Pullover scenario, as shown in Fig. 5. In both scenarios, the TOR was provided on a straight section of the highway.

First, in the Lane change scenario, there was a stopped vehicle in the right lane and the participant received the TOR warning message to change to the left lane to avoid a crash with a time budget of 10 s (i.e., 305 m from the stopped vehicle) and continued driving manually. A 10-second time budget was selected for a safe takeover based on the study by Wan and Wu (2018), which observed that crash rate after takeover was significantly low when the time budget was equal or longer than 10 s. They found that drivers did not have enough time to respond to TOR safely when the time budget was shorter than 10 s.

For the conventional TOR, only a visual warning message “Change Lane” was displayed in the windshield (i.e., the middle simulator screen) with the auditory warning from the driving simulator. For the Integrated TOR, the same visual warning message was displayed in both windshield and tablet as shown in Fig. 6 with auditory warning from both driving simulator and tablet. The volume of the tablet video was controlled to be the same for all participants and the participants were unable to change the volume by themselves. The volumes of auditory warning at the participant’s position were 60–65 dB and 78–80 dB for the conventional TOR and the integrated TOR, respectively. However, since the difference in volume was small, higher volume of auditorial alert for the integrated TOR is not likely to affect the takeover time and the post-takeover performance.

In case of the integrated TOR, at the TOR time, the video was

automatically stopped and the warning message was displayed in the tablet. The participants could press “STOP” button on the tablet to turn off the warning although they were not required to do so. During the takeover situation, there was no surrounding vehicle except the lead stopped vehicle.

Second, in the Pullover scenario, the participants were required to pull over to the right shoulder when the TOR was generated. Similar to the Lane change scenario, both conventional and integrated TOR were tested for each participant. In this scenario, a visual warning message “Pull Over to Right” was displayed in the windshield and the tablet. However, there was no surrounding vehicle during the takeover situation.

The decision-making procedure after TOR during the two scenarios was different. During the Lane change scenario, once the TOR was generated, the participants were required to analyze the driving environment. They need to make sure that they do not crash with the lead vehicle and there are no vehicles in the target lane to safely change the lane. But during the Pullover scenario, since the participants were only required to pull over the vehicle to the right shoulder, they only need to check that there is no risk of rear-end collision with any following vehicle. Due to additional tasks, the participants need longer takeover time during the Lane change scenario compared to the Pullover scenario.

The participants were instructed to put the tablet on their lap before their manual control when the TOR was initiated. In the Lane change scenario after takeover, the participants were instructed to continue driving and after about 500 m of manual driving, the scenario ended.

Each participant conducted four driving scenarios as shown in Table 1 (either wearing headphone or not wearing headphone). The participants were informed that they would test these four scenarios, but they were not informed of the order of the scenarios. The order of the scenarios was randomly assigned to each participant to reduce the learning effect. The duration of each scenario was approximately 8–12 min.

After testing each scenario, the participants were asked to answer the National Aeronautics and Space Administration Task Load Index (NASA TLX) questionnaire to assess their mental workload after the TOR warning was provided. At the end of the experiment, the participants answered a questionnaire related to their preference and experience of automated driving and TOR.

3.4. Experiment design

The research adopted a $2 \times 2 \times 2$ between- and within-subjects mixed factor experimental design. Dependent and independent variables used in the experimental design are shown in Table 2 (a) and (b).

The post-takeover performance was also evaluated to test Hypothesis 2.2 based on various driving parameters extracted from the driving simulator. These driving parameters include the minimum TTC, the standard deviation of steering wheel angle, the time to complete lane change after manual control (TCLC), and the maximum resultant acceleration during the time interval from the time of TOR warning to the



Fig. 6. Visual TOR warning messages in Lane change scenario.

Table 1

Experiment Scenarios.

Scenario	Type of TOR warning (source of visual and auditory warning)	Warning message	Wearing Headphone	Scenario name
1	Conventional (simulator only)	Change Lane	Yes	CH
2	Integrated (both simulator and tablet)	Change Lane	No	CNH
3	Conventional (simulator only)	Pullover to Right	Yes	IH
4	Integrated (both simulator and tablet)	Pullover to Right	No	INH

Table 2
Variables in experimental design.

(a) Independent variables			
Within-subjects independent variables	Level	Between-subjects independent variables	Level
TOR	Integrated TOR (I) Conventional TOR (C)	Headphones	Wearing headphone (H)
Scenario type	Lane Change Scenario Pullover Scenario		Not wearing headphone (NH)
(b) Dependent Variables			
Dependent Variables			Unit
Takeover time			s
Time to collision (TTC)			s
Time to complete lane change after manual control (TCLC)			s
Standard deviation of steering wheel angle			degree
Maximum acceleration and deceleration			m/s ²
Subjective mental workload measured by NASA TLX			1–100 (range)
(c) Post-takeover performance parameters			
Driving parameters	Definition		
Time to collision (TTC)	Remaining time until a collision occurs if both lead and following vehicles keep the same speed		
Time to complete lane change after manual control (TCLC)	Difference between the time when the manual control starts and the time when the center of the vehicle reaches the center of the left lane or the target lane (Lane change scenario only)		
Standard deviation of steering wheel angle	Variation in steering wheel angle during the takeover process		
Maximum acceleration and deceleration	Maximum acceleration and deceleration during the takeover process in the Lane change scenario and the Pullover scenario, respectively		

end time of experiment. The definitions of these post-takeover performance parameters are shown in Table 2(c). These parameters are explained in more details as follows:

TTC has been used as a measure of rear-end collision risk between the lead and following vehicles. A difference in TTC represents meaningful changes in rear-end collision risk only when the possibility of a collision is present as indicated by TTC equal to or shorter than the safety threshold (e.g., 2 s). It is expected that drivers with better post-takeover performance will recognize the surrounding traffic condition earlier and change lanes more safely. Thus, they will change lane faster to avoid a collision with the stopped vehicle and turn a steering wheel more gradually, which results in lower standard deviation of steering wheel angle. In this sense, a shorter TCLC and a smaller standard deviation of steering wheel angle indicate a safer transition from automated to manual driving. However, higher maximum acceleration and deceleration represent a more abrupt change in speed over time, which indicates an unstable transition from automated to manual driving.

3.5. Subjective measurement of mental workload

Mental workload and decision-making have a complex relationship. Decision-making tends to be better for moderate workload, whereas both overload and underload can deteriorate the quality of decision making (Soria-Oliver et al., 2017). After the TOR is generated, taking over the vehicle control is a task that requires a high amplitude of workload. Thus, a reduced mental workload will have a positive effect on the post-takeover performance.

In this regard, the NASA TLX was used to provide a reference of the subjective mental workload from different tasks (NASA, 2020). The NASA TLX measures the subjective mental workload using the six different subscales. The first three subscales relate to the demand of task,

which is mental, physical, and temporal demands; the rest of the three subscales relates to impact from the task, which is performance, effort, and frustration. It asks the user to provide a separate subjective rating based on these subscales. The participants completed the NASA TLX questionnaire using the tablet.

3.6. Preference and experience for automated driving and TOR warning

The survey was conducted to ask the participants about their experience and preference for automated driving and TOR warning as shown in Fig. 7.

3.7. Generalized linear models

Various driving performance parameters during the post-takeover situation were extracted from the driving simulator using Python and R programming language. The data from the participants who crashed or missed the TOR warning were excluded from the analysis. This is because in these cases, the participants could not complete manual control and their takeover time and post-takeover performance could not be observed.

Statistical analysis was performed using Statistical Analysis Software (SAS) version 9.4. Generalized linear models (GLM) were developed to identify the relationship between a dependent variable (takeover time, TTC, TCLC, standard deviation of steering wheel angle and the subjective mental workload value from NASA TLX) and independent variables (type of TOR warning (integrated or conventional), type of scenario (Lane change or Pullover), the use of headphone, age, gender and driving experience).

GLM was used because the model builds a linear relationship between the independent and dependent variables and it can fit to any distribution of the dependent variable, not only the normal distribution. The model can also analyze effects of continuous and categorical variables on a discrete or continuous dependent variable. The scenarios when the participants crashed (Lane change scenario) or missed the TOR completely (Pullover scenario) were excluded. In particular, the functional form of GLM ensures that the dependent variable is a non-negative value as follows:

$$\ln(y) = a + \sum_i b_i x_i$$

where y = dependent variable, x_i = independent variable i , a = constant, and b_i = coefficient for independent variable i . Since the takeover time, the post-takeover performance parameters, and the subject mental workload are non-negative values, GLM is suitable for predicting these as dependent variables.

4. Results

4.1. Descriptive statistics

Table 3 shows the descriptive statistics of the takeover time, the post-takeover performance parameters, and the subjective mental workload in different scenarios. The significance of the difference in mean of each variable between the integrated and conventional TOR was assessed using two-sample t-tests at a 95% confidence interval. Table 3(a) shows that the integrated TOR has significantly shorter mean takeover time than the conventional TOR with or without the headphone in both scenario types (p-value < 0.05), except for the case without the headphone in the Pullover scenario (p-value = 0.29). This is similar to the findings in Melcher et al. (2015) and Politis et al. (2017).

Table 3(b) compares TTC between the conventional and integrated TOR for the Lane change scenario only, because of the absence of a lead vehicle in the Pullover scenario. The table shows that the average value of minimum TTC was significantly longer for the integrated TOR than the conventional TOR with or without the headphone (p-value < 0.05).

1. Were you able to read the warning from the tablet? (Yes/No)
2. Were you able to read the warning from the windshield? (Yes/No)
3. Which warning system would you prefer in your vehicle? (Conventional/PPD Integrated)
4. How comfortable did you feel during the conventional warning system? (From 1-5 scale)
5. How comfortable did you feel during the PPD integrated warning system? (From 1-5 scale)
6. Do you think higher time budget is required for safer takeover process? (Yes/No)
7. How safe did you feel during the conventional warning system? (From 1-5 scale)
8. How safe did you feel during the PPD integrated warning system? (From 1-5 scale)
9. Are you likely to use headphone in automated driving? (Yes/No)

Fig. 7. Survey questionnaire.

Table 3

Descriptive statistics of the takeover time, the post-takeover performance parameters, and the subjective mental workload.

(a) Takeover time									
Scenario	Lane change scenario				Scenario	Pullover scenario			
(No. of obs.)	Mean	Std	Min	Max	(No. of obs.)	Mean	Std	Min	Max
CH (14)	5.20*	1.45	2.73	8.37	CH (18)	3.24*	0.93	1.60	5.72
IH (21)	3.00*	0.73	1.97	4.65	IH (20)	2.49*	0.67	1.55	3.97
CNH (18)	3.92*	1.44	1.72	6.85	CNH (21)	2.91	0.68	1.67	4.10
INH (22)	2.61*	0.59	1.48	3.76	INH (23)	2.68	0.75	1.37	4.61
(b) TTC (second) (lane change scenario only)									
Scenario (No. of observation)	Mean		Std		Min		Max		
CH (14)	2.13*		1.40		0.65		5.23		
IH (21)	4.50*		1.04		2.49		6.31		
CNH (18)	3.28*		1.61		0.71		5.89		
INH (22)	5.05*		0.90		3.51		6.69		
(c) Standard deviation of steering wheel angle (degree)									
Scenario	Lane change scenario				Scenario	Pullover scenario			
(No. of obs.)	Mean	Std	Min	Max	(No. of obs.)	Mean	Std	Min	Max
CH (14)	16.52*	16.63	0.97	54.44	CH (18)	10.62	11.23	2.59	53.60
IH (21)	8.24*	6.99	3.22	28.68	IH (20)	8.51	6.70	1.47	27.52
CNH (18)	7.55	5.49	1.70	20.58	CNH (21)	8.16*	3.27	3.20	14.24
INH (22)	6.61	3.92	1.02	17.48	INH (23)	5.64*	2.52	1.94	9.95
(d) Time to complete lane change after manual control (TCLC) (second) (Lane change scenario only)									
Scenario (No. of observation)	Mean		Std		Min		Max		
CH (14)	7.44*		1.34		4.87		9.43		
IH (21)	4.79*		0.92		3.28		6.47		
CNH (18)	5.86*		1.64		3.23		9.33		
INH (22)	4.45*		1.18		1.93		7.20		
(e) Subjective mental workload (range: 1–100)									
Scenario	Lane change scenario				Scenario	Pullover scenario			
(No. of obs.)	Mean	Std	Min	Max	(No. of obs.)	Mean	Std	Min	Max
CH (14)	44.24*	24.39	1.67	92.67	CH (18)	27.39	13.95	7.67	51.00
IH (21)	28.90*	15.91	5.00	61.67	IH (20)	22.90	21.56	2.33	82.67
CNH (18)	42.54	20.29	5.00	70.33	CNH (21)	32.21*	17.34	6.67	71.00
INH (22)	31.27	16.21	2.33	66.67	INH (23)	21.51*	14.22	2.33	46.33

*Means are statistically different between the integrated and conventional TOR at a 95% confidence interval.

The number of cases when the minimum TTC is shorter than the safety threshold (e.g., 2 s) is an important indicator of driver safety. The threshold of 2 s was determined based on a minimum risk of collision set by Brown et al. (1991). A total of 18 participants (11 participants with headphones and 7 participants without headphones) showed TTC shorter than 2 s in the conventional TOR, but no participant showed TTC shorter than 2 s in the integrated TOR. Thus, the integrated TOR did not

only increase the minimum TTC, but also reduced the frequency of safety critical events.

Tables 3(c) shows that the mean of standard deviation of steering wheel angle was significantly smaller for the integrated TOR than the conventional TOR in the case with the headphone in the Lane change scenario and the case without the headphone in the Pullover scenario (p -value < 0.05).

Table 3(d) shows that the mean TCLC was significantly smaller for the integrated TOR than the conventional TOR with or without the headphone in the Lane change scenario (p -value < 0.05). However, means of maximum acceleration were not statistically different between the integrated and conventional TOR for all scenarios. Table 3(e) shows that the integrated TOR had significantly higher mean subjective mental workload than the conventional TOR in the case with the headphone in the Lane change scenario and the case without the headphone in the Pullover scenario (p -value < 0.05).

No participants crashed during the scenarios with the integrated TOR (IH and INH). But with the conventional TOR, participants crashed 12 times (8 with headphones and 4 without headphones) in the Lane change scenario and 7 participants missed the TOR warning (5 with headphones and 2 without headphones) in the Pullover scenario. This indicates that the integrated TOR effectively reduced the likelihood of crash and missing the TOR warning.

4.2. Components of visual and physical readiness time

Table 4 shows that the components of visual and physical readiness time measured using force-sensitive sensors and cameras. Due to a small number of observations (i.e., 6 to 8 observations for each scenario), the differences in the components between the conventional and integrated TOR were assessed using two non-parametric statistical tests (Mann-Whitney U test and bootstrapping test). The differences in medians (Mann-Whitney U test) and means (bootstrapping test) between the conventional and integrated TOR were assessed at a 95% confidence interval.

The results of both tests show that when the headphone was not used, the Clearance Time was significantly longer for the integrated TOR than the conventional TOR in the Lane change scenario at a 95% confidence interval. This is mainly because the additional time was required to read the warning from the tablet in the integrated TOR.

Also, the result of the bootstrapping test shows that when the headphone was used, the Readiness Time was significantly shorter for

Table 4

Mean physical and visual readiness time components (seconds) during takeover process between conventional and integrated TOR.

(a) Lane change scenario				
Scenario name (No. of observation)	TOR Gaze Redirection Time (Std)	Clearance Time (Std)	Maneuver Gaze Redirection Time (Std)	Readiness Time (Std)
CH (7)	1.83 (0.59)*	1.35 (0.60)	0.26 (0.27)	1.00 (0.55)
IH (7)	–	1.71 (0.43) **	0.40 (0.31)	0.56 (0.31)
CNH (6)	1.29 (0.34)*	1.11 (0.27)	0.38 (0.21)	0.54 (0.27)
INH (8)	–	1.61 (0.42) **	0.35 (0.26)	0.43 (0.30)
(b) Pullover scenario				
Scenario name (No. of observation)	TOR Gaze Redirection Time (Std)	Clearance Time (Std)	Maneuver Gaze Redirection Time (Std)	Readiness Time (Std)
CH (7)	1.45 (0.32)*	1.66 (0.53)	0.38 (0.32)	0.55 (0.27)
IH (7)	–	1.69 (0.33) **	0.53 (0.34)	0.61 (0.32)
CNH (7)	1.57 (0.50)*	1.56 (0.66)	0.50 (0.33)	0.43 (0.20)
INH (6)	–	1.65 (0.59) **	0.46 (0.23)	0.72 (0.20)

*The TOR Gaze Redirection Time includes the TOR Recognition Time for the conventional TOR.

**The Clearance Time includes the TOR Recognition Time for the integrated TOR.

Bold indicates that median or mean values are statistically different between the conventional and integrated TOR at a 95% confidence interval.

the integrated TOR than the conventional TOR in the Lane change scenario at a 95% confidence interval. On the other hand, the Readiness Time was rather longer for the integrated TOR in the Pullover scenario. This indicates that the integrated TOR helped drivers take control faster in more urgent condition when the driver intervention (i.e., lane change to avoid a collision with the lead vehicle) is immediately required. However, due to absence of the TOR Gaze Redirection Time as shown in Fig. 4, the takeover process was faster for the integrated TOR than the conventional TOR in both scenarios.

4.3. Generalized linear models

GLMs were estimated to identify the relationship between the takeover time and the related factors as shown in Table 5(a). GLMs were also estimated to identify the relationship between the post-takeover performance parameters and the related factors as shown in Table 5(b). In both scenarios, the effects of the integrated TOR were significant at a 95% confidence interval (p -value < 0.05). However, the goodness-of-fit of GLMs for the Pullover scenario was low ($R^2 = 0.09 - 0.11$) compared to the Lane change scenario.

4.4. Preference and experience for automated driving and TOR warning

It was found that 87% of the participants would prefer the integrated TOR over the conventional TOR. Also, the survey indicated that 80% of participants would prefer the time budget longer than 10 s.

A majority (68%) of the participants reported maximum level of comfort for the integrated TOR, whereas only 4% of the participants reported the same level of comfort for the conventional TOR. Similarly, a majority (75%) of the participants reported maximum level of safety for the integrated TOR, but only 11% of the participants reported the same level of safety for conventional TOR. The significance of difference in mean levels of comfort and safety between the integrated and conventional TOR was also assessed using two-sample t -tests. It was found that the mean level of comfort was significantly higher for the integrated TOR (4.6) than the conventional TOR (3.2) at a 95% confidence interval (p -value < 0.05). Similarly, the mean level of safety was significantly

Table 5

Estimated parameters of generalized linear models.

(a) Takeover time					
Scenario	Lane Change	Pullover			
R ²	0.55	0.09			
Error degrees of freedom	96	98			
Constant	0.69 (<.0001)	1.08 (<.0001)			
Integrated TOR	−0.18* (<.0001)**	−0.17 (0.0013)			
Headphone	0.07 (0.0029)	—			
Male	−0.08 (0.0004)	—			
Driving Experience	−0.01 (0.0009)	—			
(b) Post-takeover performance					
	Lane change scenario		Pullover scenario		
	TTC	Std of steering wheel angle	Lane change duration	Std of steering wheel angle	Max deceleration
R ²	0.39	0.12	0.38	0.09	0.11
Constant	0.94 (<.0001)	1.94 (<.0001)	1.79 (<.0001)	2.09 (<.0001)	1.83 (<.0001)
Integrated TOR	0.70 (<.0001)	−0.37 (0.0171)	−0.34 (<.0001)	−0.36 (0.0020)	0.15 (0.0114)
Headphone	−0.23 (0.013)	0.43 (0.0051)	0.14 (0.0038)	—	—
Gender	—	—	—	—	—
Age	—	—	—	—	−0.20 (0.0092)

*Coefficient of independent variable

**Numbers in parentheses are p -values.

higher for the integrated TOR (4.5) than the conventional TOR (2.9) at a 95% confidence interval (p -value < 0.05).

5. Discussion

Based on the comparison of components of visual and physical readiness time, the reason for why the integrated TOR can reduce the takeover time was identified as follows. This is because the integrated TOR can save the time it takes for the drivers to redirect their gaze to read the TOR warning message on the dashboard or windshield, not necessarily because it can reduce both visual readiness time and physical readiness time. However, the integrated TOR did not consistently reduce the times it took for the drivers to check the TOR warning and be physically ready for manual driving in both Lane change and Pullover scenarios. Thus, the Hypotheses 1.1 and 1.2 are rejected.

Based on the result of GLMs, the factors associated with the takeover time were discussed as follows. In the Lane change scenario, the takeover time was relatively shorter for male participants than female participants. This result is consistent with [Lipps et al. \(2011\)](#), which found that male drivers generally showed shorter reaction time than female drivers. Also, the takeover time was shorter for the participants with longer driving experience. This result is consistent with the fact that the drivers with longer driving experience have quicker driving-related reflexes. For instance, [Wright et al. \(2016\)](#) found that more experienced middle-aged drivers visually identified hazardous situations quicker compared to less experienced drivers.

The use of headphones increased the takeover time. This result is similar to the study by [de Waard et al. \(2011\)](#), which found that listening to music using earbuds during cycling increased the cyclists' response time to the auditory information. On the other hand, this is not the expected effect for the integrated TOR because the drivers can hear louder and more salient auditory TOR warning from headphones. However, the interaction effect of the integrated TOR and the headphone use on the takeover time was not significant. This indicates that the auditory TOR warning from PPD through headphones in the integrated TOR did not significantly reduce the time for TOR recognition and visual and physical readiness. This finding contradicts the hypothesis that the use of headphones will reduce drivers' cognitive distractions and help them recognize the TOR warning faster. Thus, Hypothesis 2.1 is rejected. This is potentially because the drivers could still recognize the warning faster even without using headphones as the video was automatically stopped and the visual TOR warning was displayed on the tablet.

In the Pullover scenario, only the integrated TOR showed significant negative effect on the takeover time. There were no outliers of the takeover time for the Pullover scenario. The R^2 value of the model (= 0.09) was low because the impact of the TOR warning on the takeover time was relatively less significant when the drivers were not required to urgently react to avoid a crash and the takeover situation was relatively safer.

Based on the result of GLMs, the factors associated with the post-takeover performance were also discussed as follows. In the Lane change scenario, the integrated TOR significantly increased TTC. This indicates that the integrated TOR can increase the safety margin to avoid a collision. The integrated TOR also decreased standard deviation of steering wheel angle and TCLC, which represent better post-takeover performance.

However, the use of headphones significantly reduced TTC while increasing the standard deviation of steering wheel angle and TCLC. This indicates that the use of headphones generally decreased the post-takeover performance for both integrated and conventional TOR. This is similar to the findings in [de Waard et al. \(2011\)](#). Moreover, there was no significant interaction effect of the type of TOR warning and the use of headphones. This indicates that the auditory TOR warning from PPD through headphones did not significantly improve the post-takeover performance. Thus, Hypothesis 2.2 is rejected.

The result also shows that in the Pullover scenario, the integrated

TOR significantly reduced the standard deviation of steering wheel angle but increased the maximum deceleration. Also, the maximum deceleration was lower for older drivers. Overall, the signs of coefficients of the 'Integrated TOR' variable in the GLMs indicate that the integrated TOR generally improved post-takeover performance compared to the conventional TOR. [Politis et al. \(2017\)](#) also found positive effect of the integrated TOR on the post-takeover performance – the reduction in lateral deviation after takeover.

GLMs were also estimated for the subjective mental workload but the model fit was very low ($R^2 = 0.04$ for the Lane change scenario and $R^2 = 0.09$ for the Pullover scenario). Due to an insignificant difference in the effects on the subjective mental workload between the integrated and conventional TOR, Hypothesis 3 is rejected.

Lastly, the survey shows the participants' preference and experience for automated driving and TOR warning. Based on their responses, the participants generally felt higher levels of comfort and safety level with the integrated TOR than the conventional TOR.

6. Conclusions and recommendations

This study investigated the effects of integrating takeover request (TOR) warning with a personal portable device (PPD) called the "integrated TOR" on takeover time and post-takeover performance in level 3 automated driving by comparing with the conventional TOR warning without a PPD. Main findings of this study are summarized as follows:

First, the integrated TOR significantly reduced average takeover time compared to the conventional TOR. It was found from the force-sensitive resistor and camera data that the integrated TOR could reduce the takeover time mainly because it eliminated the time required to redirect the driver's gaze to read the TOR warning message on the dashboard or windshield. Moreover, as the video was automatically stopped (i.e., the drivers could not perform the secondary task) after the TOR warning was generated, the drivers could respond to the TOR faster.

Second, the participants generally showed better post-takeover performance with the integrated TOR than the conventional TOR. The integrated TOR increased the minimum time-to-collision (TTC) and decreased the time to complete lane change after manual control (TCLC) in the Lane Change scenario where the participants were requested to change lane to avoid a collision with the stopped vehicle. Also, the integrated TOR decreased the standard deviation of steering wheel angle in both Lane Change scenario and Pullover scenario where the participants were requested to pull over to the right shoulder. As a result, there was no crash with the integrated TOR unlike the conventional TOR.

Third, the use of headphones during the secondary task generally increased the takeover time and decreased the post-takeover performance in the Lane change scenario. Thus, the auditory TOR warning from PPD in the integrated TOR did not effectively reduce the time for TOR recognition and visual and physical readiness. This is potentially because the drivers could recognize the visual TOR warning from the tablet faster even without headphones after the video was automatically stopped.

Fourth, the impacts of TOR warning on the takeover time and the post takeover-performance were more significant in more urgent takeover situation (Lane change scenario) than less urgent takeover situation (Pullover scenario).

Fifth, the average value of subjective mental workload was lower for the integrated TOR than the conventional TOR for both Lane change and Pullover scenarios. This shows that the integrated TOR can improve drivers' comfort during the takeover process.

Lastly, the participants preferred the integrated TOR over the conventional TOR due to higher subjective safety and comfort levels.

In summary, a unique finding in this study is that the integrated TOR in PPD can significantly reduce the takeover time and improve the post-takeover performance in both urgent and less urgent conditions. However, the finding shows that the use of headphones for this integrated TOR did not significantly reduce the takeover time or improve the post-

takeover performance. Due to this limitation and the restriction of headphone use for Level 3 driving automation, headphone use is not recommended for listening to auditorial warning from PPD.

However, the study has several limitations. First, the difference in the results between the integrated and conventional TOR might have occurred not only due to different location of TOR, but also different visual design of the alerts (e.g., color and size of text) between PPD and vehicle interface. Thus, sole effect of location of TOR could not be identified in this study. Second, there is a limitation of ecological validity of the driving simulator scenario. For instance, the design speed of 110 km/h, which was selected based on a typical speed limit on a limited-access highway in Canada, may be too high for safe takeover of SAE Level 3 vehicles. Thus, it is recommended to conduct the study using lower operating speed to simulate more realistic conditions for takeover. Third, most participants were in a younger age group and there were more male participants than female participants. Lastly, the study could not observe the materials the participants watched on the tablet, which could have affected their level of concentration on the secondary task and their reaction to the TOR warning.

In future studies, it is recommended to conduct the experiment for participants with a wider range of age, driving experience, and equal number of male and female participants for generalized results. It is also recommended to evaluate the effectiveness of the integrated TOR in different road geometry, traffic, and weather conditions while participants engage in different types of secondary tasks.

CRedit authorship contribution statement

Niloy Talukder: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chris Lee:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Yong Hoon Kim:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Balakumar Balasingam:** Writing – review & editing, Investigation, Conceptualization. **Francesco Biondi:** Writing – review & editing, Investigation, Conceptualization. **Aditya Subramani Murugan:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **Eunsik Kim:** Writing – review & editing, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Niloy Talukder is a Research Assistant of Civil and Environmental Engineering at the University of Windsor, Canada. He received the Master's degree in civil engineering from the University of Windsor. Mr. Talukder has conducted research on the development of warning strategies for partially-automated driving.

Chris Lee, Ph.D., P.Eng. is a Professor of Civil and Environmental Engineering at the University of Windsor, Canada. He received the Ph.D. in civil engineering from the University of Waterloo, Canada. His research interests include vehicle interaction and conflicts in car-truck mixed traffic and manual-autonomous vehicle mixed traffic, car-following and lane-changing behaviour, driver safety in partially-automated vehicles, advanced real-time speed control for freeways, and aggressive and distracted driving.

Yong Hoon Kim, Ph.D., P.Eng. is an Associate Professor of Civil and Environmental Engineering at the University of Windsor, Canada. He received the Ph.D. in civil engineering from Purdue University, U.S.A. His research interests include advanced driver assistance systems, connected and autonomous vehicles, intelligent transportation systems, advanced traveler information systems, transportation system analysis and modeling, and traffic flow modeling and simulation.

Balakumar Balasingam, Ph.D., P.Eng. is an Associate Professor of Electrical and Computer Engineering at the University of Windsor, Canada. He received the PhD in electrical engineering from McMaster University, Canada. His research interests include autonomous (cyber, physical and human) systems, signal processing, machine learning, and information fusion.

Francesco Biondi, Ph.D. is an Associate Professor of Kinesiology at the University of Windsor, Canada. He received the Ph.D. in experimental psychology from the University of Padova, Italy. His research interests include human factors in manufacturing and transportation, workload assessment, and user-centered design.

Aditya Subramani Murugan is a Ph.D. student of Industrial Engineering at the University of Windsor, Canada. His research interests include computer vision, neural networks, and gamification.

Eunsik Kim, Ph.D., P.Eng. is an Assistant Professor of Industrial Engineering at the University of Windsor, Canada. He received the Ph.D. in industrial engineering from Pennsylvania State University, U.S.A. His research interests include occupational ergonomics, ergonomic intervention, physiological measurement and analysis, biomechanics, musculoskeletal disorders, and manual material handling.